University of Cape Town

Department of Electrical Engineering

EEE3094S - Control Systems Engineering

Problem Set 4.2: Rough Root Locus Solutions

Question 1

For systems with the poles and zeros described,

i) Sketch a rough root locus for the system (as in, just the general shape)

ii) Is it possible for the closed-loop system to become unstable?

iii) Is it possible for the closed-loop response to become oscillatory?

a) Poles at $-2 \pm 3j$

b) Poles at -5 and -6, zeros at -2 and 1

c) Pole at -4 and zero at 2

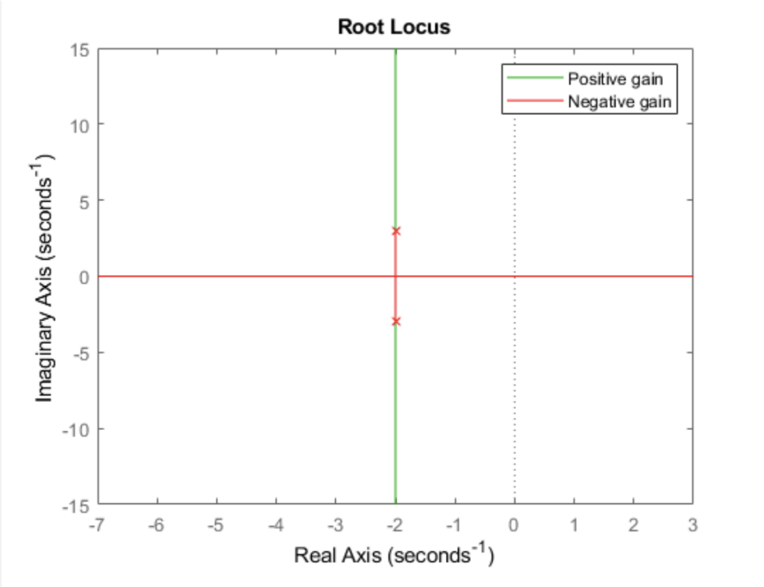
d) Poles at -3 and -1, zeros at $1 \pm j$

e) Poles at $-1 \pm 5j$, zero at -5

Solutions

a) Poles at $-2 \pm 3j$

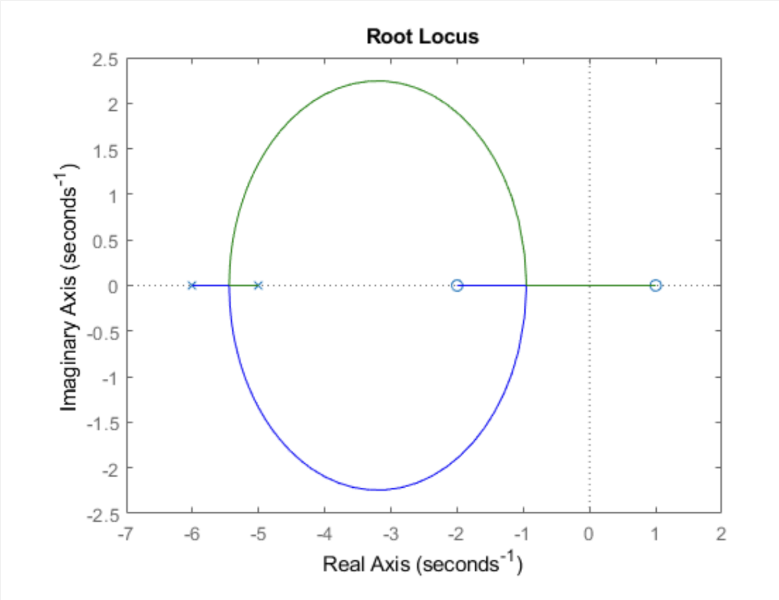
Solution:
i)



ii) Yes, at low gain the system will be unstable
iii) Yes, it already is

b) Poles at -5 and -6, zeros at -2 and 1

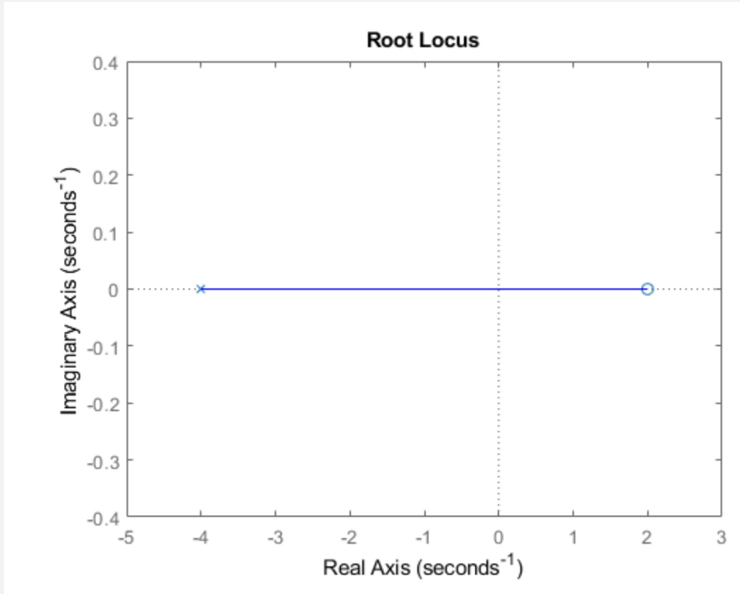
Solution:
i)



ii) Yes, as gain tends to infinity
iii) Yes, for a specific range of gain values

c) Pole at -4 and zero at 2

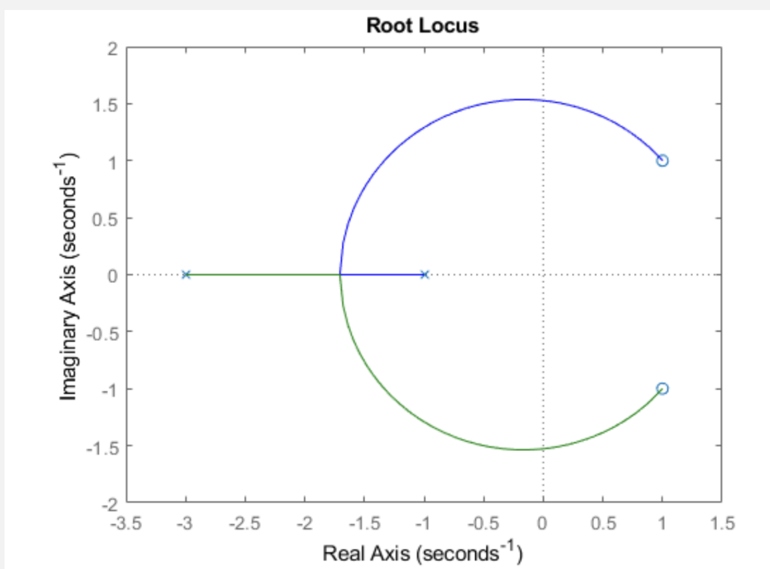
Solution:
i)



ii) Yes, as gain tends to infinity
iii) No, there is only one pole so it can't become oscillatory

d) Poles at -3 and -1, zeros at $1 \pm j$

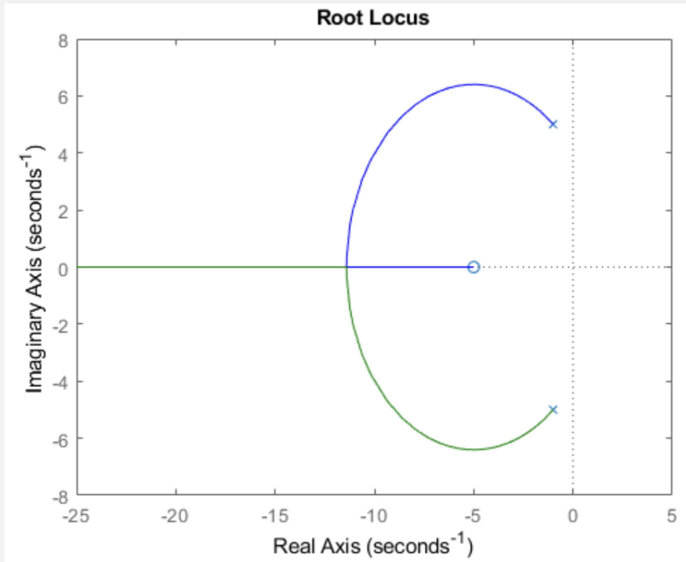
Solution:
i)



ii) Yes, as gain tends to infinity
iii) Yes, as gain increases

e) Poles at $-1 \pm 5j$, zero at -5

Solution:
i)



ii) Not for increasing gain values
iii) Yes, but they stop with increasing gain values